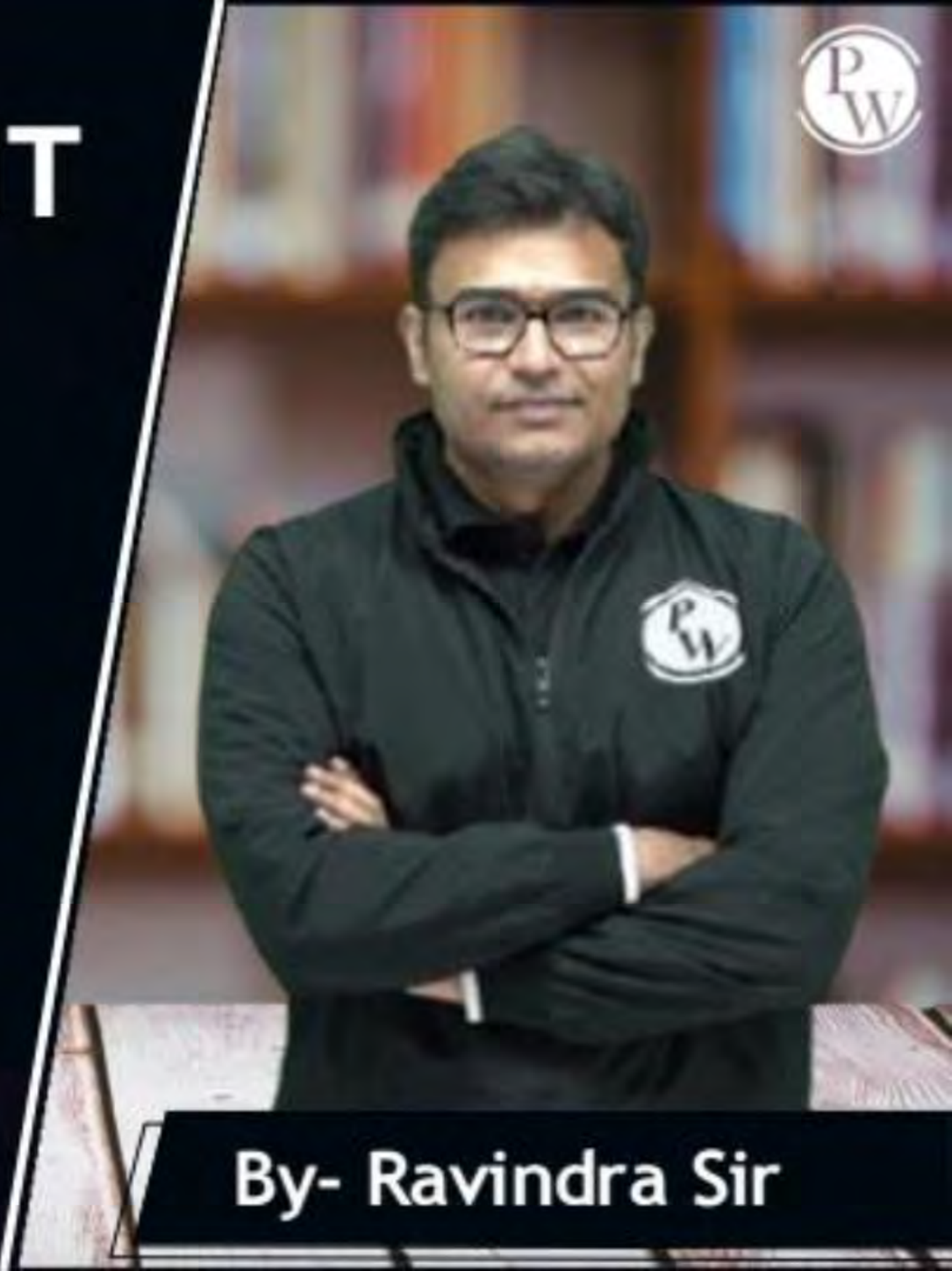
Computer Science & IT

COMPILER DESIGN
(CD)

CO

Lecture No. 3



By- Ravindra Sir

Recap of Previous Lecture



- Topic CO
- Topic
- Topic

Topics to be Covered



- Topic CO
- Topic
- Topic

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Inspiring Stories : Mula Gabharu

Mula Gabharu – The Queen Who Raised Women Soldiers

Background: Mula Gabharu was the daughter of an Ahom king in Assam in the early 1500s.

Struggles: Her husband fell fighting an invading army. Her people were helpless and losing hope.

Achievements: She trained women warriors, led them into battle with courage, and confronted the enemy herself before being martyred.

Impact: If she could turn grief into action and lead women into battle, you can turn doubt into study and lead your rank up.



Inspiring Stories : B. N. Jagatap

B. N. Jagatap

Background: Grew up in a modest family in rural Maharashtra with no strong science background at home.

Education: PhD in Nuclear Magnetic Resonance (NMR) from IISc Bangalore.

Career Achievements: Improved NMR methods to study molecules in great detail, helping drug design and materials research.

Impact: Made high-end science tools more effective for Indian research labs and universities.



Inspiring Stories : Veeramangai Kuyili

Veeramangai Kuyili – The Woman Who Became a Human Torch

Background: Kuyili was a trusted soldier of queen Velu Nachiyar from Tamil Nadu in the 1700s.

Struggles: She worked without praise and lived in a time when women were rarely seen in battle.

Achievements: During one attack, she covered her body in ghee, set herself on fire, and jumped into the British armory. Her act became a key turning point in the war.

Impact: If she could light herself on fire to save her queen, you can light up that answer sheet with focus



Inspiring Stories : K. J. Rao

K. J. Rao

Background: Born in a small town in Karnataka with very limited means. Had to work hard to afford studies.

Education: PhD in Solid State Chemistry from IISc Bangalore.

Career Achievements: Developed new ceramic materials that can store energy and work in extreme heat.

Impact: Helped India make better energy storage systems and heat-resistant tools for industry.

machine independent optimizations:-

1) Constant folding

```
int a = 10 X 5
int b = 100/2
int c = 3 + 7*2
```

⇓ folding

```
int a = 50
int b = 50
int c = 17
```

2) Constant propagation :-

```
int x=5
int y = x+3
int z = y*2
```

⇓ CP

```
int x=5
int y=8
int z=16
```

3) Common Subexpression Elimination :-

```
int a = (x+y) * 2;
int b = (x+y) * 3;
```

CSE

```
int temp = x+y;
int a = temp * 2
int b = temp * 3
```

4) Dead Code elimination :-

```
int a=10;
int b=20;
a=30;
int c=a+b;
```

```
int a=30;
int b=20;
int c=a+b
```

```
if (false)
{
   X
}
```

Algebraic Simplification :-

```
int a = x*1;
int b = y+0;
int c = z-0;
```

after Simplification

```
int a = x
int b = y
int c = z
```

Tail recursion elimination:-

```
int factorial (int n, int result)
{  if (n==0) return result;
   return factorial(n-1, result*n));
}
```

After tail recursion elimination:-

```
int factorial (int n)
{
  while(n!=0)
  {
    result = result *n;
     n = n-1;
  }
  return result;
}
```

machine dependent optimizations :-

1) Register allocation
   - local allocation (CO)
   - Global allocation
2) use of addressing modes (CO)
3)

3) Peephole optimization

a) Redundant load and store elimination

$x = y + z$

```
mov y, R0
add z, R0
mov R0, x
```

$a = b + c$

$d = a + e$

```
mov b, R0
add c, R0
mov R0, a
mov a, R0
add e, R0
mov R0, d
```

✓

b) flow of Control optimization

Avoid jumps on jumps

L_1: JMP L_2 L_4

L_2: Jump L_3

L_3: Jump L_4

eliminate dead code

```
# define x 0
if (x)
{
    dead code
}
```

c) Use of machine idioms :-

$i = i + 1$	mov i, R0 add 1, R0 mov R, i } inc i

THANK - YOU